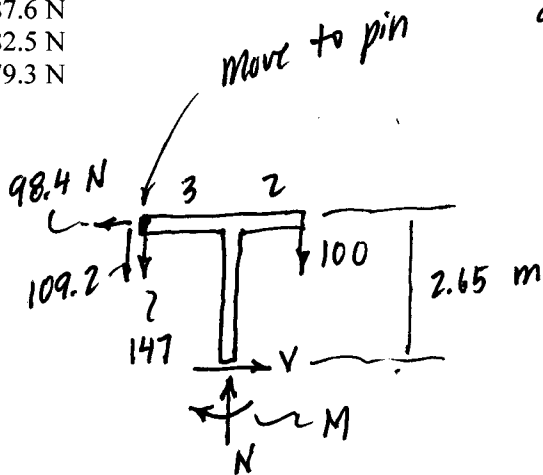


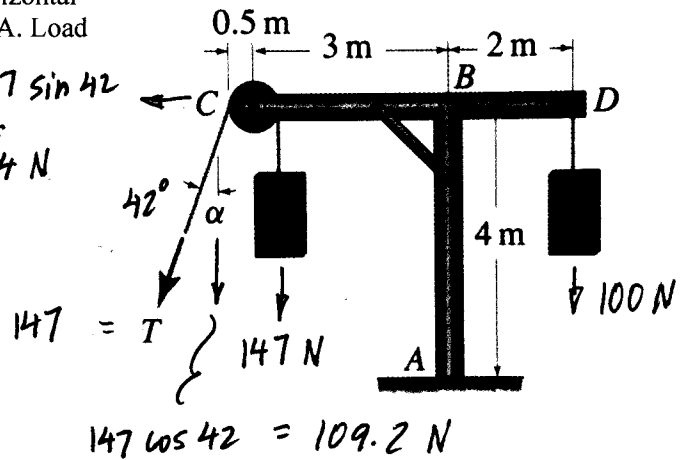
Exam 3: Practice Exam

1. (4 pt) Determine the magnitude of the shear force at a horizontal section on frame ABCD that lies 1.35 m above fixed support A. Load W is 147 N, $\alpha = 42^\circ$ and the counterweight Q is 100 N.

- A. 98.4 N
B. 92.1 N
C. 87.6 N
D. 82.5 N
E. 79.3 N



$$147 \sin 42^\circ = 98.4 \text{ N}$$



$$147 \cos 42^\circ = 109.2 \text{ N}$$

By inspection: $V = 98.4 \text{ N}$

2. (4 pt) Determine the magnitude of the normal force at a horizontal section on frame ABCD that lies 1.35 m above fixed support A. Load W is 147 N, $\alpha = 42^\circ$ and the counterweight Q is 100 N.

- A. 356 N
B. 362 N
C. 368 N
D. 374 N
E. 383 N

By inspection:

$$N = 100 + 147 + 109.2$$

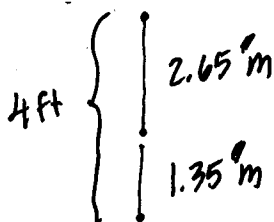
$N = 356.2 \text{ N}$ (C)

3. (5 pt) Determine the magnitude of the bending moment at a horizontal section on frame ABCD that lies 1.35 m above fixed support A. Load W is 147 N, $\alpha = 42^\circ$ and the counterweight Q is 100 N.

- A. 829 N-m
B. 838 N-m
C. 846 N-m
D. 812 N-m
E. 801 N-m

$$+\circlearrowleft \sum M_{cut} = 0; \quad M + 100(2) - 98.4(2.65) - (147 + 109.2)(3) = 0$$

$M = 829.4 \text{ N}\cdot\text{m}$



Section 2 – Shear and Moment Diagrams

Provide information for constructing the **shear diagram**. The x origin is on the left at A and the support reactions are $A_y = 6$ kip acting up, $C_y = 9$ kip acting up and $M_C = 60$ kip-ft acting clockwise. A is a roller, B is a pin and C is a fixed support.

4. (3 pt) What happens to the shear diagram at $x = 12$ ft?

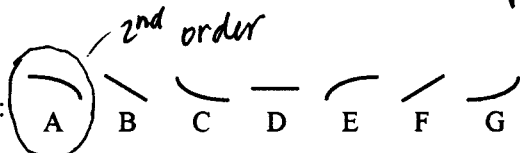
- A. No change
B. Decrease 9 kip
C. Increase 9 kip
D. Decrease 3 kip
E. Increase 3 kip
F. Decrease 1.5 kip

Determine the numerical value (magnitude only) of the change in shear for section $0 < x < 12$ ft as well as the behavior (shape) of the shear for the same region.

5. (3 pt) Change = -9 kip (magnitude only)

- A. 9
B. 1.5
C. 6
D. 12
E. 18
F. 0

6. (3 pt) Behavior:



7. (3 pt) What happens to the shear diagram at $x = 16$ ft?

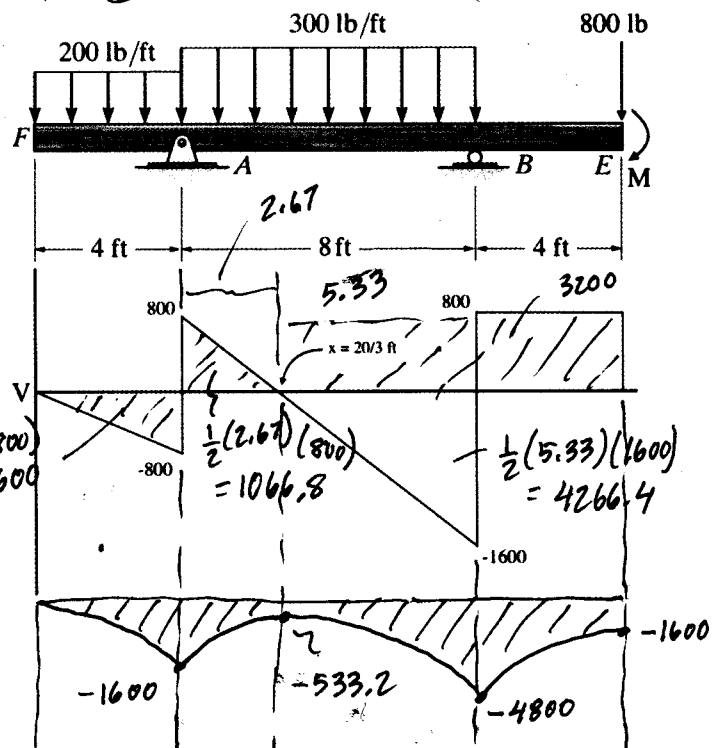
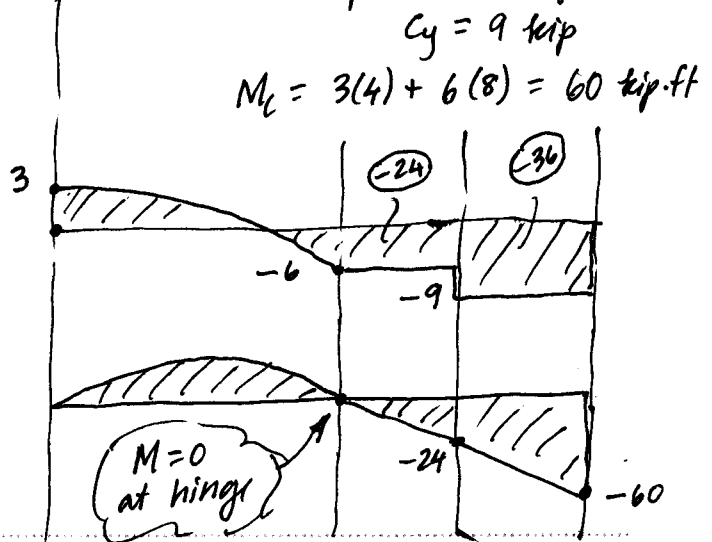
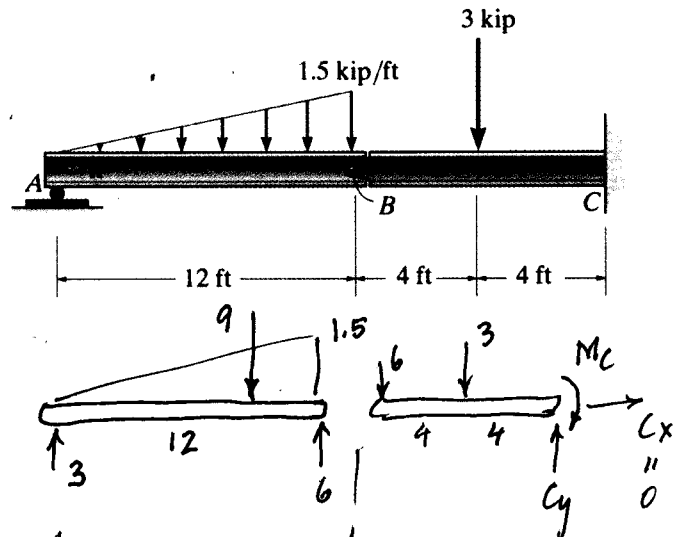
- A. Decrease 3 kip
B. Increase 3 kip
C. Decrease 9 kip
D. Increase 9 kip
E. Decrease 6 kip
F. Increase 6 kip

Provide information for constructing the **moment diagram**. The x origin is on the left at A and the support reactions are $A_y = 1600$ lb acting up, $B_y = 2400$ lb acting up. A is a pin and B is a roller. The couple $M = 1600$ lb-ft CW at E.

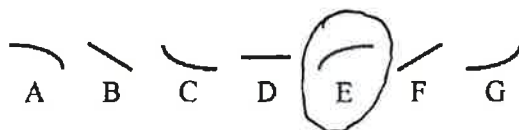
Determine the numerical value (magnitude only) of the change in moment for section $4 < x < 6.667$ ft as well as the behavior (shape) for the same region.

8. (3 pt) Change = _____ lb-ft (magnitude only)

- A. 1067
B. 800
C. 1600
D. 2400
E. 4267
F. 3200



9. (3 pt) Behavior:



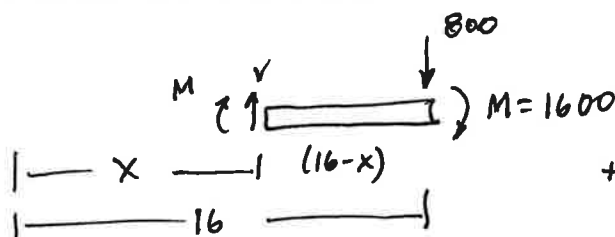
10. (3 pt) What happens to the moment diagram at $x = 16$ ft?

- A. Increase 1600 lb·ft
B. Decrease 400 lb·ft
C. Increase 400 lb·ft
D. Decrease 1600 lb·ft
E. Increase 800 lb·ft
F. No change



11. (4 pt) Determine the order of the equation for the internal bending moment for the region $12 < x < 16$ ft on the beam shown.

- A. third order, cubic
B. constant
C. second order, parabolic
D. zero
E. first order, linear



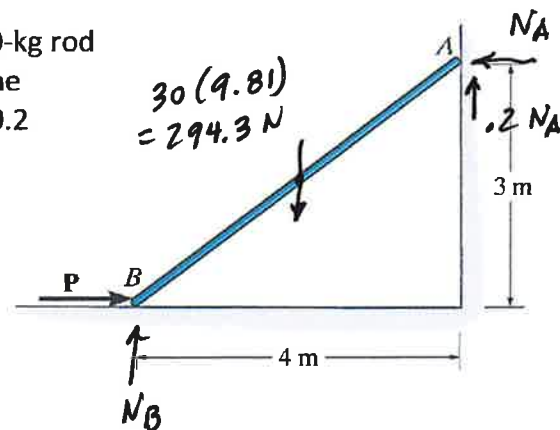
$$+\circlearrowleft M + 800(16-x) + 1600 = 0$$

$$M = -800(16-x) - 1600$$

Section 3 - Friction

12. (5 pt) Determine the minimum force P to prevent the uniform 30-kg rod AB from sliding down. The contact surface at B is smooth, whereas the coefficient of static friction between the rod and the wall at A is $\mu_s = 0.2$

- A. 143 N
B. 155 N
C. 138 N
D. 129 N
E. 117 N

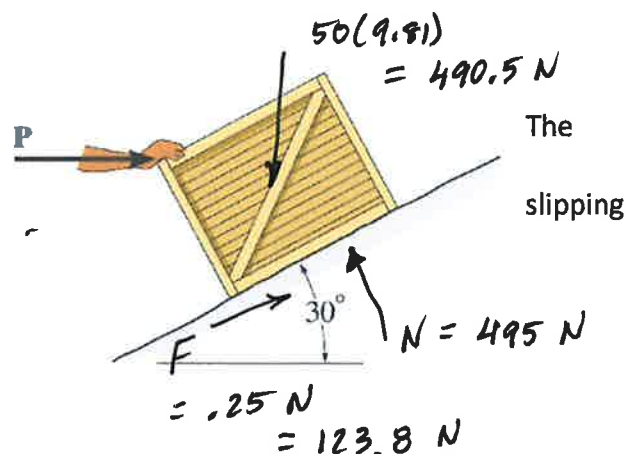


$$+\circlearrowleft \sum M_B = 0; -294.3(2) + N_A(3) + 0.2N_A(4) = 0$$

$$N_A = 154.9 \text{ N}$$

$$P = N_A$$

13. (5 pts) Determine the minimum horizontal force **P** required to prevent the crate from slipping **down** the plane. The crate has a mass of 50 kg and the coefficient of static friction between the crate and the plane is $\mu_s = 0.25$. At the point of downward, the normal force at the contact surface is 495 N.



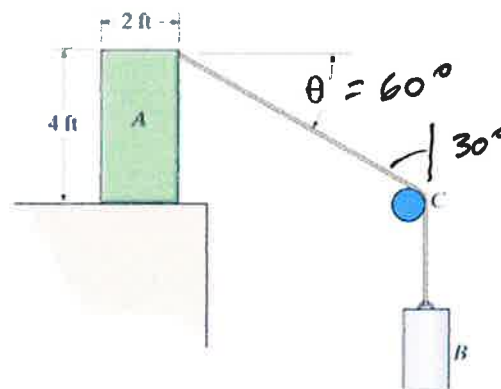
- A. 140 N
- B. 203 N
- C. 449 N
- D. 474 N
- E. 335 N

$$\begin{aligned} \rightarrow P \cos 30 - 490.5 \sin 30 \\ + 123.8 = 0 \end{aligned}$$

$$P = 140.2 \text{ N}$$

14. (5 pts) If $\theta = 60^\circ$, what is the value of β when using the belt friction equation for the cord at peg C?

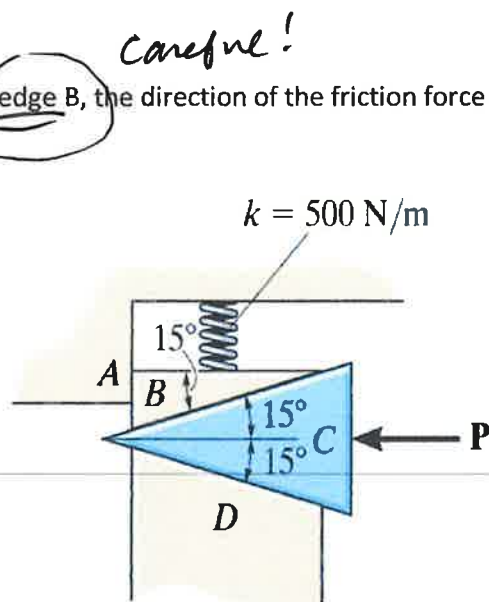
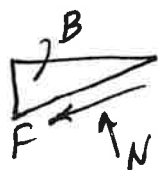
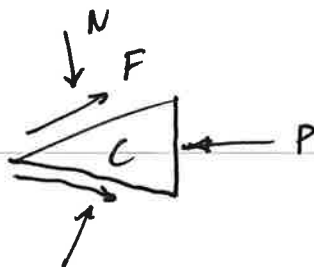
- A. 1.05 radians
- B. 0.52 radians
- C. 0.79 radians
- D. 1.12 radians
- E. 0.68 radians



$$\begin{aligned} \beta &= 30^\circ \cdot \frac{\pi}{180} = \frac{\pi}{6} = \\ &= .524 \text{ rad} \end{aligned}$$

15. Force P is applied to cause motion. On a free-body diagram of wedge B , the direction of the friction force at the contact surface between B and C would be drawn....

- A. down and to the right
- B. up and to the left
- C. down and to the left
- D. up and to the right



Section 4 – Centroids A

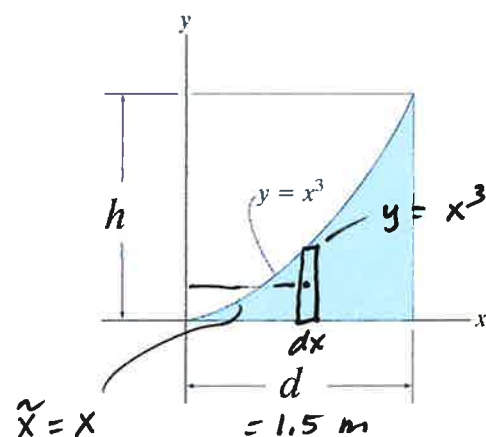
16. (5 pt) Determine the location of $\bar{x}$ of the shaded area. $d = 1.5$ m
The area of the shaded part is 1.266 m^2

- A. 1.04 m
- B. 0.96 m
- C. 1.20 m
- D. 0.83 m

$$\bar{x} \text{ Num} = \int_0^{1.5} x \cdot x^3 dx$$

$$= \int_0^{1.5} x^4 dx = \frac{1}{5} (1.5)^5 = 1.519 \text{ m}$$

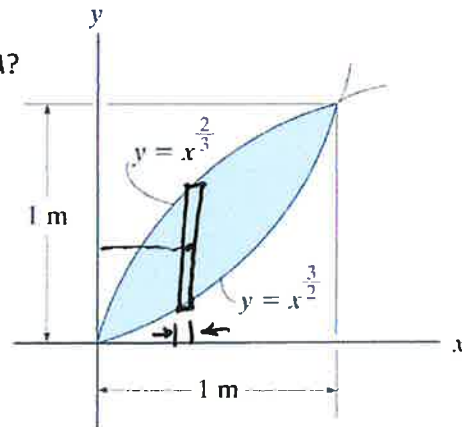
$$\bar{x} = \frac{1.519}{1.266} = 1.20 \text{ m} = \bar{x}$$



$$dA = y \cdot dx = x^3 dx$$

17. (5 pt) When using a vertical differential element of area to solve for the location of $\bar{x}$ for the shaded area what expression would be correct for dA ?

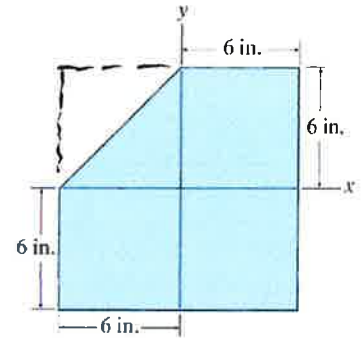
- A. $(x^{667} + x^{1.5})/2 dx$
- B. $(x^{667} - x^{1.5}) dx$
- C. $(x^{667} + x^{1.5}) dx$
- D. $(x^{667}) dx$



18. (5 pt) Determine the location of $\bar{y}$ for shaded area. Note location of the origin.

A. -0.57 ft B. -0.63 ft C. -0.75 ft D. -0.86 ft

Segment			



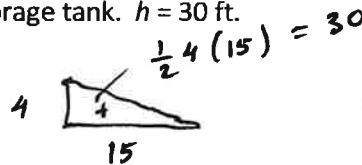
$A = 144$ (for the square) $(0,0)$
 $A = \frac{1}{2}(6)(6) = 18 \text{ in}^2$ (for the triangle)
 $(-4, 4)$ (centroid of the triangle)

$$\bar{y} = \frac{\sum \tilde{y} A_i}{\sum A_i} = \frac{0 + (4)(-18)}{144 - 18} = \frac{-72}{126} = \boxed{-.571 = \bar{y}}$$

19. (5 pt) (4 pt) Determine the **volume** of the storage tank. $h = 30$ ft.

- A. 18,614 ft³
 B. 21,012 ft³
 C. 19,803 ft³
 D. 22,148 ft³

$$\theta = 2\pi$$

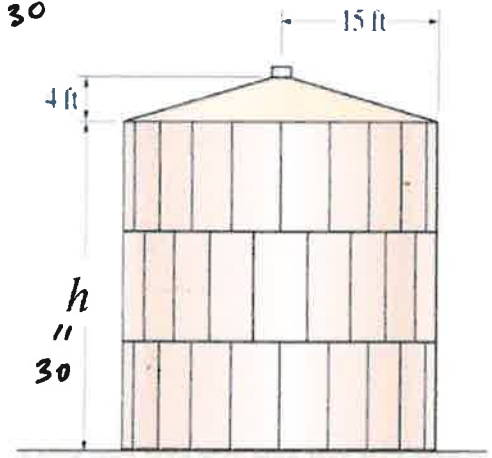
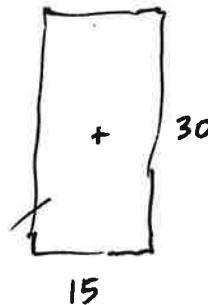


$$V = \sum \theta \bar{r} A$$

$$= 2\pi [5(30) + 7.5(450)]$$

$$\boxed{V = 22,148 \text{ ft}^3}$$

$$A = 450$$



20. (5 pt) Determine the **outside** surface area of the storage tank including the roof. $h = 30$ ft. (do **not** include the floor).

- A. 3559 ft²
 B. 3253 ft²
 C. 3197 ft²
 D. 3088 ft²

$$A = 2\pi [7.5(15.52) + 15(30)]$$

$$\boxed{A = 3559 \text{ ft}^2}$$

$$L = \sqrt{4^2 + 15^2} = 15.52 \text{ ft}$$

